

Grade 6 lessons

Ratios, percents, negatives and equations

IN THIS BOOK

1. Ratios and rates
2. Percents
3. Negative numbers
4. Expressions and equations
5. Area and data

HOW TO USE IT

Read one lesson together, then practice it. Each lesson matches 5 questions in the Grade 6 Kahoot and Blooket sets, and one stop on the Grade 6 trail in the Math Trail app.

Every lesson has the big idea, the steps, two worked examples, one mistake to watch for, and the words to know.

Ratios and rates

THE BIG IDEA

A ratio compares two amounts. A rate compares amounts with different units, like miles and hours. A unit rate tells you the amount for just 1.

HOW TO DO IT

1. Write a ratio as a to b, or a:b.
2. To make an equal ratio, multiply or divide both parts by the same number.
3. To find a unit rate, divide so that the second amount becomes 1.
4. Use the unit rate to find any other amount by multiplying.

WORKED EXAMPLES

$2:3 = \square:12$

3 was multiplied by 4 to get 12.

Multiply 2 by 4 as well: 8. So $2:3 = 8:12$.

3 notebooks cost \$6. How much is 1?

$6 \div 3 = 2$. Each notebook costs \$2.

WATCH OUT

Order matters. $2:3$ is not the same as $3:2$. And always multiply both parts. Adding the same number to both parts changes the ratio.

WORDS TO KNOW

ratio

a comparison of two amounts

unit rate

the amount for one

Percents

THE BIG IDEA

Percent means "out of 100". 25% is 25 out of 100, which is $\frac{25}{100}$ or 0.25.

HOW TO DO IT

1. To change a percent to a decimal, divide by 100. 25% becomes 0.25.
2. To find a percent of a number, multiply the number by that decimal.
3. Use shortcuts: 50% is half, 25% is a quarter, 10% is divide by 10.
4. To find what percent one number is of another, divide part by whole, then multiply by 100.

WORKED EXAMPLES

What is 25% of 80?

25% is one quarter. $80 \div 4 = 20$.

15 is what percent of 60?

$15 \div 60 = 0.25$. Multiply by 100: 25%.

WATCH OUT

0.5 is 50%, not 5%. Moving from a decimal to a percent shifts the point two places.

WORDS TO KNOW

percent

out of one hundred

Negative numbers

THE BIG IDEA

Negative numbers are less than zero. On a number line they sit to the left of zero. The further left, the smaller the number.

HOW TO DO IT

1. Picture a number line. Adding a positive moves right. Subtracting moves left.
2. Adding a negative moves left: $4 + (-10)$ is the same as $4 - 10$.
3. Subtracting a negative moves right: $5 - (-3)$ is the same as $5 + 3$.
4. To compare, the number further right is greater.

WORKED EXAMPLES

$$4 + (-10) = ?$$

Start at 4. Move 10 to the left. You land on -6.

$$5 - (-3) = ?$$

Subtracting a negative is adding. $5 + 3 = 8$.

WATCH OUT

-8 looks bigger than -3 but it is smaller, because it is further left on the number line.

WORDS TO KNOW

negative

less than zero

integer

a whole number, positive, negative or zero

Expressions and equations

THE BIG IDEA

In algebra a letter stands for a number. Solving an equation means finding the number that makes it true.

HOW TO DO IT

1. To work out an expression, replace the letter with its number, then calculate.
2. To solve an equation, undo what was done to the letter.
3. Whatever you do to one side, do the same to the other side.
4. Check by putting your answer back into the equation.

WORKED EXAMPLES

If $x = 4$, what is $3x + 2$?

$3 \times 4 = 12$. Then $12 + 2 = 14$.

Solve $x + 7 = 15$.

Subtract 7 from both sides. $x = 8$.

Check: $8 + 7 = 15$.

WATCH OUT

$3x$ means 3 times x . If x is 4, then $3x$ is 12, not 34.

WORDS TO KNOW

variable

a letter that stands for a number

equation

two things joined by an equals sign

Area and data

THE BIG IDEA

A triangle is half of a rectangle, so its area is half of base $\times$ height. For a list of numbers, the mean, median and range each describe it in one number.

HOW TO DO IT

1. Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height}$.
2. Mean: add all the numbers, then divide by how many there are.
3. Median: put the numbers in order and take the middle one.
4. Range: biggest number minus smallest number.

WORKED EXAMPLES

A triangle has base 10 and height 6. What is its area?

$$\frac{1}{2} \times 10 \times 6 = 30.$$

Find the mean and median of 3, 9, 5, 1, 7.

Mean: $3 + 9 + 5 + 1 + 7 = 25$, and $25 \div 5 = 5$. Median: in order they are 1, 3, 5, 7, 9. The middle one is 5.

WATCH OUT

Put the numbers in order before you find the median. The middle of the list as written is usually wrong.

WORDS TO KNOW

mean

the average

median

the middle value

range

the spread from smallest to biggest